

JOY EKKA

PhD Candidate in Computational Chemistry & Materials Modelling

 joyekka.tcmp@gmail.com

Professional Summary: Computational materials scientist (MSc Physics; PhD in Theoretical Chemistry & Modelling, expected Dec 2026) specializing in multi-scale modelling of porous materials. Strong command of quantum, atomistic and statistical methods like DFT, GCMC, and MD applied to predicting adsorption, transport, and stability of MOF/polymer composites under industrially relevant conditions. Alongside my PhD, I am actively expanding into AI & machine learning for material properties and discovery by working through hands-on projects on graph neural networks and bayesian active learning, generative design approaches and bringing a first-principles understanding of chemistry & physics to data-driven materials modelling.

Research Experience

PhD Researcher - MOF/Polymer Composites for VOCs Capture 2023 - present
CNRS / Université de Montpellier & CNES, France

- Developed an integrated multi-scale computational workflow (DFT, GCMC, MD) to design and characterize MOFs/hydrophobic polymer composites for selective Volatile organic compounds (VOCs) capture under humid conditions relevant to spacecraft environments
- Computational predictions directly guided experimental fabrication and testing of composite materials, establishing an iterative computation-experiment feedback loop

Visiting Researcher Sep - Nov 2022
Universität Duisburg-Essen, Germany

- Electronic structure studies of complex materials using DFT methods

Master's Researcher – Electronic structure and properties of 2D materials 2020 - 2021
Theoretical Condensed Matter Physics & Advanced Computational Materials Science Lab, IIT Indore

- DFT investigation of Mn-doped MoS₂ monolayers as electrocatalysts for hydrogen evolution; work published in Physical Chemistry Chemical Physics (RSC, 2022)

Publications

- MOF/Hydrophobic Polymer Composites for Toluene Capture. <https://doi.org/10.1021/acsami.6c09571>
- Unveiling the Role of 2D Monolayer Mn-doped MoS₂: Toward an Efficient Electrocatalyst for H₂ Evolution Reaction. <https://doi.org/10.1039/d1cp04344g>
- 1 paper under review.

Education

PhD in Theoretical Chemistry & Computational Modelling 2023 – Dec 2026
CNRS / Université de Montpellier, France | Co-funded: CNES & CNRS
Thesis: Metal-Organic Framework/hydrophobic polymer composites for effective capture of contaminants in the space domain
Supervisors: Prof. Guillaume Maurin, Dr. Sabine Devautour-Vinot, Dr. Guillaume Rioland (CNES)

MSc in Physics 2019 – 2021
Indian Institute of Technology (IIT) Indore, India

BSc (Honours) in Physics 2016 – 2019
St. Xavier's College, Ranchi, India

Conferences & Presentations

- COMET-STR 2026: Oral presentation
- E-MRS Spring Meeting 2026: Oral presentation
- EUROMOF 2025: Poster presentation
- CNES Annual Research Meeting JC2 2025: Oral & poster presentation, Toulouse
- CNES Annual Research Meeting JC1 2024: Oral & poster presentation, Toulouse
- MOFSIM 2024: Attendee

Technical Skills

Modelling methods: Molecular Dynamics (MD), Density Functional Theory (DFT), Grand Canonical Monte Carlo (GCMC), Graph theory, Numerical Computing, Physics-based modelling.

Softwares: LAMMPS, GROMACS, VASP, CP2K, Crystal, Gaussian, RASPA, VMD, VESTA, Materials Studio, ASE.

Programming & Scientific Computing: Python, C++, MATLAB, NumPy/SciPy, Linux/HPC environments, Workflow Automation.

Machine Learning (developing): PyTorch, scikit-learn; learning graph neural networks and Bayesian active learning for materials property prediction; familiar with generative materials models.

Languages: English (fluent), Hindi (native), French (basic - A2)